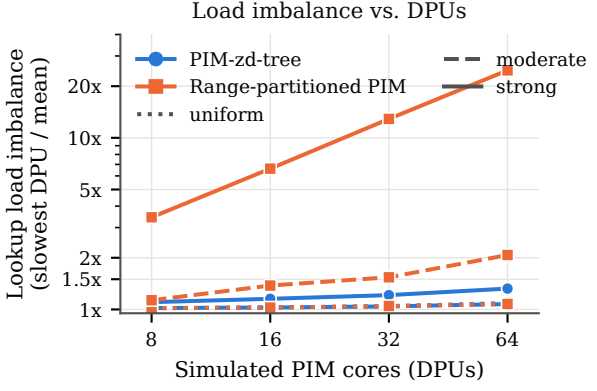
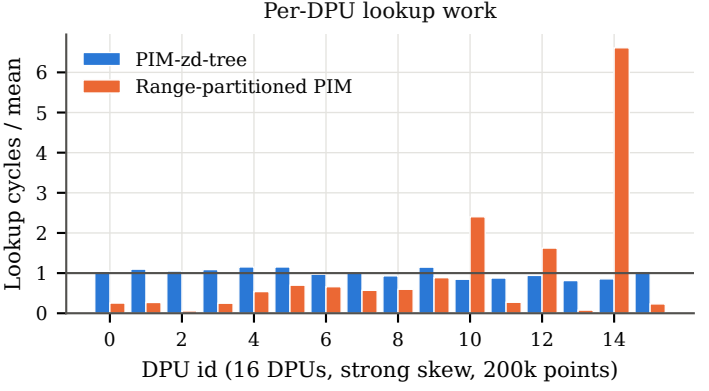

Load imbalance vs. DPUs
PIM-zd-tree
Range-partitioned PIM
uniform
moderate
strong
Lookup load imbalance
(slowest DPU / mean)
20x
10x
5x
2x
1.5x
1x
8
16
32
64
Simulated PIM cores (DPUs)


Per-DPU lookup work
PIM-zd-tree
Range-partitioned PIM
Lookup cycles / mean
6
5
4
3
2
1
0
0
2
4
6
8
10
12
14
DPU id (16 DPUs, strong skew, 200k points)